A Modular Quantitative Trading Pipeline:
Alpha Generation, Reinforcement Learning Control, and Risk-Aware Backtesting

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Sequential Decision-Making Project

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Abstract

This paper documents a modular quantitative trading pipeline implemented in `quant_stack`, a module within the `stock_trading` package. The system combines cross-sectional factor investing, cointegration-based statistical arbitrage, volatility forecasting, regime detection, and a lightweight recurrent forecast sleeve with reinforcement-learning-based portfolio allocation, hedging, and execution components. The design is intentionally hybrid: classical quantitative finance supplies the alpha and risk features, while reinforcement learning is used as a sequential control layer for exposure, overlay intensity, and tail-risk management. On the current backtest window, the full pipeline attains equity-like return with a materially improved drawdown profile relative to SPY, while remaining close to the strongest factor-only benchmark in upside capture.

Keywords: quantitative trading, reinforcement learning, factor investing, statistical arbitrage, regime detection, risk management, backtesting

1 Introduction

Many quantitative trading systems are built from a collection of specialized modules rather than from a single monolithic predictor. Cross-sectional factors, relative-value signals, volatility models, and regime filters often contribute distinct pieces of information about expected return and risk. Reinforcement learning (RL), by contrast, is most compelling when the task is inherently sequential: choosing when to increase risk, when to hedge, and how to adapt execution to changing conditions [8, 1].

This project adopts exactly that separation of responsibilities. The alpha layer is intentionally finance-first. It draws on ideas from factor investing [5, 7], cointegration-based statistical arbitrage [4], conditional volatility modeling [3, 2], and regime detection [6]. The RL layer does not attempt to discover alpha from scratch. Instead, it treats alpha as an input and learns how aggressively to express it under different risk and market states.

The main contributions of the implementation are fourfold.

1. A modular code architecture that cleanly separates data, alpha, control, backtesting, and visualization concerns.
2. A factor-anchored portfolio construction policy in which RL primarily controls exposure and active overlay size rather than acting as a second stock selector.
3. A light dynamic hedging sleeve that delivers convex-protection-like behavior without requiring a full options-chain pipeline.
4. A research-style reporting workflow that connects theory, architecture, and empirical diagnostics in a reproducible form.

From a research perspective, this framing matters because it narrows the gap between a teaching prototype and a system that can be discussed in the language of quantitative portfolio construction. The objective is not to claim that the pipeline is production-ready. Rather, it is to assemble a well-motivated stack in which each modeling choice has a clear financial role and where performance can be decomposed into alpha generation, risk transformation, and control.

2 Related Literature and Design Motivation

The report is not meant to claim novelty in any individual modeling choice. Instead, it builds a coherent pipeline from well-established ingredients.

Factor investing. The cross-sectional component is aligned with the idea that common risk factors help explain returns [5]. Momentum is motivated by the medium-horizon continuation evidence of Jegadeesh and Titman [7], while quality and low-volatility proxies are included to stabilize the cross-section.

Statistical arbitrage. The pairs sleeve follows the classical cointegration logic of Engle and Granger [4]: if a spread is mean reverting around an equilibrium relation, deviations from that equilibrium may carry predictive content.

Volatility and regimes. Conditional heteroskedasticity motivates the GARCH sleeve [3, 2], while the market-regime component uses a two-state hidden Markov model in the spirit of Hamilton [6].

Reinforcement learning. RL is used as a controller rather than a raw alpha generator. This follows the practical observation that in finance, RL is often most useful for allocation, inventory, and execution decisions rather than for replacing domain-specific signal construction [8, 1].

3 System Architecture

The pipeline is organized as a linear sequence of stages, illustrated in Figure 1. Each stage has a clearly delimited responsibility: upstream stages produce signals and risk features, the control layer translates these into portfolio decisions, and the backtest layer evaluates realized outcomes.

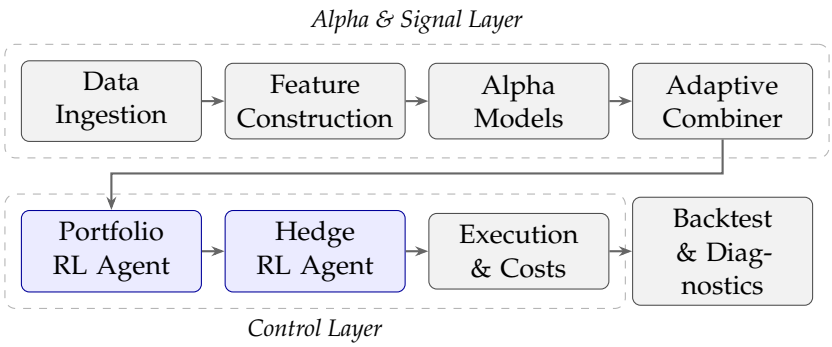


Figure 1: High-level architecture of the quantitative trading pipeline. The alpha and signal layer (top) produces expected-return scores and risk features; the control layer (bottom) translates these into portfolio weights, hedge intensity, and cash allocation before evaluation in the backtest engine.

3.1 Software Structure

The code is split into the following modules:

Table 1: Module-level decomposition of the `quant_stack` package.

Module	Responsibility
`config.py`	Universe definition, global constants, and external series identifiers
`data.py`	Market price/volume ingestion, macro series (FRED / BLS fallback), and SEC company facts
`alpha.py`	Cross-sectional factors, pairs cointegration, GARCH volatility, HMM regime, LSTM forecast, and adaptive signal combination
`rl.py`	Portfolio construction agent, execution controller, and dynamic hedging agent
`pipeline.py`	Walk-forward backtest orchestration and online RL update loop
`plots.py`	Equity-curve, drawdown, RL behavior, and alpha-model diagnostic figures
`main.py`	CLI entry point and compatibility wrapper

This decomposition is important for research reproducibility. It keeps the external command stable while turning the internals into separately testable components.

4 Data and Feature Construction

4.1 Universe

The traded universe spans four broad buckets. The growth sleeve comprises AAPL, MSFT, GOOGL, AMZN, META, and NVDA. Financials and energy are represented by JPM, GS, XOM, and CVX.

The defensive sleeve includes JNJ, PG, KO, PEP, and WMT, while diversifiers are captured through GLD, TLT, XLU, and VNQ. SPY serves as the passive equity benchmark throughout.

4.2 Data Sources

The pipeline supports both a free baseline configuration and a richer research configuration. Daily prices and volumes are sourced via `yfinance`. When a `FRED_API_KEY` is available, macro data are drawn directly from FRED; in its absence, the system falls back to BLS and U.S. Treasury Fiscal Data. SEC company facts are incorporated when `SEC_USER_AGENT` is set, enabling a light fundamentals layer without requiring a proprietary data subscription.

4.3 Macro Regime Inputs

When available, the macro sleeve includes rates and labor-market variables such as the 10-year yield, 2-year yield, federal funds rate, unemployment rate, and term spread. These variables do not form a standalone alpha book. Instead, they help create a macro regime belief that is blended with the HMM regime estimate.

5 Methodology

5.1 Problem Formulation

Let $r_{t+1} \in \mathbb{R}^N$ denote next-period asset returns for a universe of N tradable assets. At time t, the system observes a feature state $x_t \in \mathcal{X}$ constructed from cross-sectional factor values, relative-value spreads, volatility forecasts, regime beliefs, and macro inputs. The alpha layer maps these features into an expected-return score vector $\alpha_t \in \mathbb{R}^N$ and a confidence vector $c_t \in \mathbb{R}^N$.

The control layer then chooses a portfolio allocation w_t, a cash allocation, and a hedge intensity h_t. The one-step portfolio objective is not written as a pure return maximization problem. Instead, it is implicitly multi-objective: maximize long-run wealth while limiting drawdown, volatility, and unnecessary turnover. This is the reason the project separates alpha estimation from control: the two tasks operate on different time scales and carry different inductive biases.

5.2 Cross-Sectional Factor Model

The factor sleeve combines four z-scored components: momentum, a value proxy, a quality proxy, and a low-volatility tilt. Letting m_i, v_i, q_i, and ℓ_i denote the standardized scores for asset i, the composite factor score is

$$f_i = 0.30m_i + 0.25v_i + 0.25q_i + 0.20\ell_i.$$

This choice reflects a balanced, multi-style factor book rather than a single directional bet.

5.3 Pairs Trading

For candidate pairs, a hedge ratio β is estimated by OLS and the spread

$$s_t = P_t^{(1)} - \beta P_t^{(2)}$$

is normalized into a z-score

$$z_t = \frac{s_t - \mu_s}{\sigma_s + \varepsilon}.$$

Large positive z-scores correspond to short-spread trades and large negative z-scores correspond to long-spread trades.

5.4 Volatility and Regime Models

Each name receives a GARCH(1,1) forecast

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

which is used for confidence scaling and risk awareness. Separately, a two-state HMM produces a bull-versus-bear regime belief. The final regime signal is

$$\text{RegimeBelief}_t = 0.75 \cdot \text{HMMBelief}_t + 0.25 \cdot \text{MacroBelief}_t.$$

5.5 Adaptive Alpha Combination

Let $s_{i,t}^{(k)}$ denote the signal from source k for asset i at time t. The expected return score is formed as

$$\alpha_{i,t} = \sum_k w_t^{(k)} s_{i,t}^{(k)},$$

where the source weights $w_t^{(k)}$ adapt over time based on recent realized signal quality. In the implementation, this is proxied by recent Spearman information coefficients between source scores and realized returns.

This adaptive weighting is important because the constituent alpha sleeves are not equally informative across all market states. Pairs signals can be weak in strongly trending markets, deep nonlinear forecasts can degrade when regime structure shifts, and factor signals can vary in efficacy depending on the dispersion of the cross-section. The adaptive combiner therefore behaves like a simple online model-selection layer.

5.6 Portfolio Construction RL

The portfolio RL state summarizes three inputs: the current cross-sectional alpha opportunity, recent realized portfolio volatility, and the prevailing regime belief. Crucially, the policy is *factor-anchored*. The target book is

$$w_t^{\text{target}} = 0.90\, w_t^{\text{factor}} + 0.10\, w_t^{\text{stabilizer}},$$

where the stabilizer is a long-only minimum-variance-style portfolio estimated from a Ledoit–Wolf shrinkage covariance matrix.

If the RL agent chooses invested fraction b_t and active overlay size τ_t, the final invested portfolio is approximately

$$w_t = b_t \left[(1 - \tau_t) w_t^{\text{target}} + \tau_t w_t^{\text{satellite}} \right].$$

The satellite sleeve is not a fully independent book; it tilts the factor target toward the strongest alpha names. This keeps the full pipeline close to the working factor engine while still letting RL modulate aggressiveness.

5.7 Reward Design

The portfolio RL update uses a differential Sharpe-style reward based on the realized portfolio return relative to a recent rolling distribution of returns. This means the agent is not rewarded solely for taking more risk in absolute terms; it is rewarded for improving return relative to the recent volatility environment. The hedge agent uses an asymmetric realized-return reward in which losses are penalized more heavily than gains are rewarded. That asymmetry is a deliberate design choice: it pushes the hedge layer toward preserving capital in stress rather than maximizing average carry.

5.8 No-Trade Band and Hedging

To reduce turnover drag, a rebalance band suppresses small changes in target weights. When only part of the portfolio is changed, the active weights are rescaled to preserve the desired capital budget.

The hedging RL layer observes drawdown, volatility regime, and recent momentum. It chooses one of four hedge intensities. The total one-period return can be written schematically as

$$r_t^{\text{total}} = r_t^{\text{risky}}(1 - \phi h_t) + r_t^{\text{cash}} + r_t^{\text{hedge}},$$

where h_t is the hedge fraction and r_t^{hedge} includes both carry cost and a convex stress payoff approximation.

Although stylized, this formulation captures the intended economics of a protection sleeve: hedging should be mildly costly in calm periods, increasingly helpful in volatile or negative markets, and most valuable when drawdowns and adverse momentum coincide.

6 Experimental Design

The backtest uses a walk-forward protocol over approximately five years of daily data. The sample is split evenly into training and test segments. The LSTM sleeve is first trained on the initial window, after which the full pipeline is rolled forward one trading day at a time.

At each date, the pipeline executes the following sequence. First, factor, pairs, GARCH, HMM, macro, and LSTM features are generated from the available data. These are combined into expected-return scores and confidence levels via the adaptive combiner. The portfolio RL agent then selects an exposure bucket and overlay size, after which the hedging agent independently chooses the hedge intensity. Transaction costs, cash carry, and hedge payoffs are applied to arrive at the realized portfolio return, which is finally used to update both RL value tables before advancing to the next trading day.

The comparison set includes SPY buy-and-hold, an equal-weight benchmark, and a factor-only strategy with no RL overlay.

6.1 Benchmark Logic

The benchmark set is chosen to isolate different sources of value. SPY measures whether the pipeline adds value relative to a standard passive equity exposure. Equal weight tests whether the broadened universe alone produces useful diversification benefits. Factor-only isolates the pure

alpha layer without any RL overlay. Comparing the full pipeline against factor-only is particularly important because it reveals whether RL is adding value as a controller or merely acting as a drag on a strong stock-selection engine.

6.2 Transaction Costs and Carry

The backtest includes a simple turnover-based transaction-cost model and daily risk-free carry on uninvested cash. These assumptions remain stylized, but they are sufficient to force the allocator to confront a meaningful trade-off between responsiveness and trading friction. The no-trade band is best understood in this context: it is not just an implementation convenience, but a structural attempt to reduce noise trading in a daily-rebalanced system.

7 Evaluation Metrics

The reporting suite includes annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, and return-distribution diagnostics including a daily 5% Value-at-Risk estimate.

If $\{r_t\}_{t=1}^{T}$ denotes daily portfolio returns and r_f is the annual risk-free rate, then the annualized Sharpe ratio is

$$\text{Sharpe} = \frac{252\,\bar{r} - r_f}{\sqrt{252}\,\hat{\sigma}(r) + \varepsilon}.$$

The Sortino ratio replaces total volatility with downside volatility, while the Calmar ratio scales annualized return by the magnitude of maximum drawdown. Each metric emphasizes a different aspect of the return distribution, which matters in this project because the pipeline is explicitly designed to reshape tail risk, not merely to maximize mean return.

8 Results

The most recent strong run uses a test window from September 20, 2023 to March 23, 2026. The summary metrics are shown in Table 2.

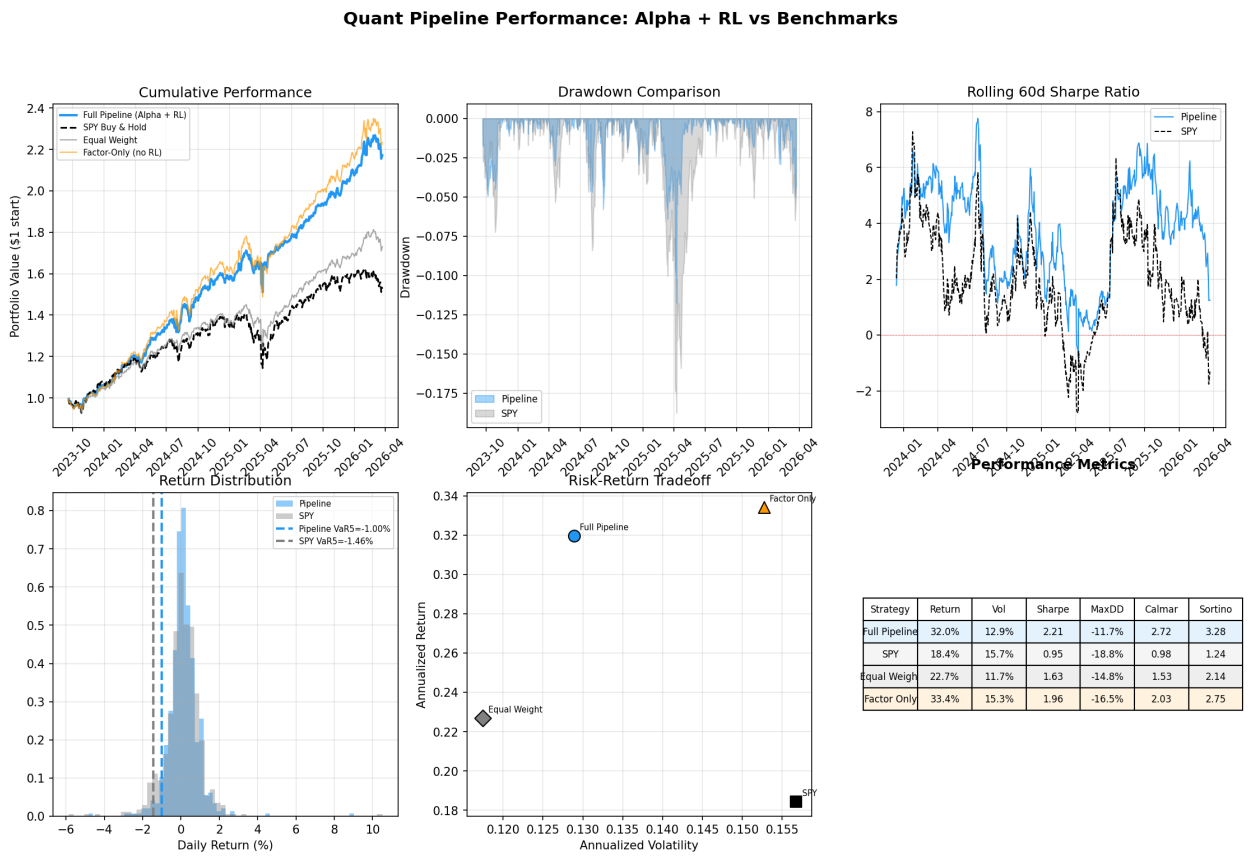
Table 2: Performance summary on the current walk-forward test window.

Strategy	Return	Volatility	Sharpe	Max DD	Calmar
Full Pipeline	32.0%	12.9%	2.21	-11.7%	2.72
SPY	18.4%	15.7%	0.95	-18.8%	0.98
Equal Weight	22.7%	11.7%	1.63	-14.8%	1.53
Factor Only	33.4%	15.3%	1.96	-16.5%	2.03

The key interpretation is that the full pipeline retains most of the upside of the factor-only strategy while materially reducing drawdown and volatility. Relative to SPY, the pipeline improves both return and risk-adjusted performance. Relative to the factor-only benchmark, it gives up a small amount of raw return in exchange for better risk control.

The resulting profile is informative. The factor-only strategy remains the most aggressive expression of the alpha book, which is consistent with its higher volatility and deeper drawdown. The full pipeline, by contrast, sits closer to what one would expect from an institutional overlay: it captures much of the equity upside, preserves a strong Sharpe ratio, and converts some of the raw return advantage into a shallower left tail.

Figures 3–5 provide further behavioral detail. The RL diagnostics (Figure 3) confirm that both the portfolio and hedge agents vary their actions in a regime-consistent manner rather than converging to a fixed policy. The alpha model diagnostics (Figure 4) illustrate how the adaptive combiner shifts weight toward sleeves with higher recent information coefficients. The execution RL panel (Figure 5) shows that the no-trade band meaningfully reduces turnover relative to unconstrained daily rebalancing, with the cost savings concentrated in low-alpha-conviction periods.



Strategy	Return	Vol	Sharpe	MaxDD	Calmar	Sortino
Full Pipeline	32.0%	12.9%	2.21	-11.7%	2.72	3.28
SPY	18.4%	15.7%	0.95	-18.8%	0.98	1.24
Equal Weigh	22.7%	11.7%	1.63	-14.8%	1.53	2.14
Factor Only	33.4%	15.3%	1.96	-16.5%	2.03	2.75

Figure 2: Performance comparison for the full pipeline and benchmark strategies.

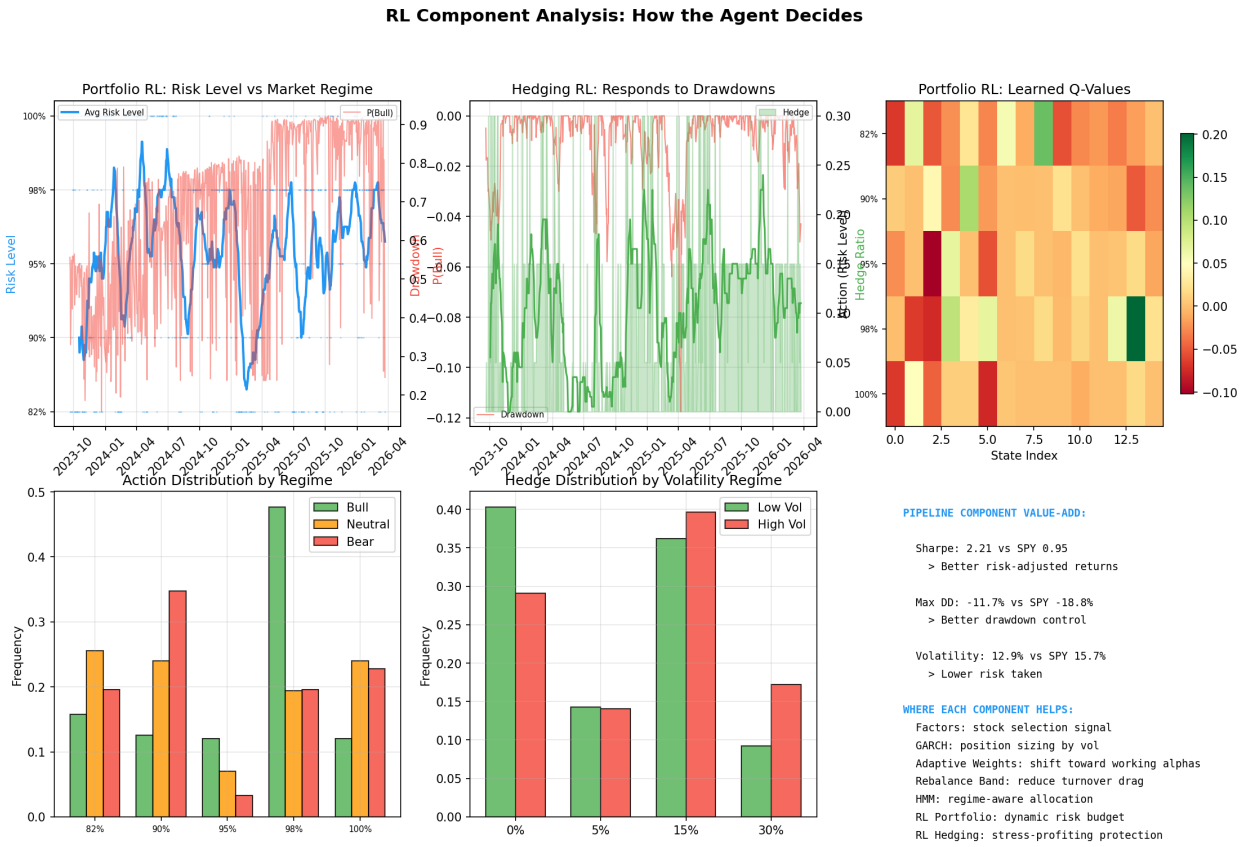


Figure 3: Behavioral diagnostics for the portfolio and hedging RL layers.

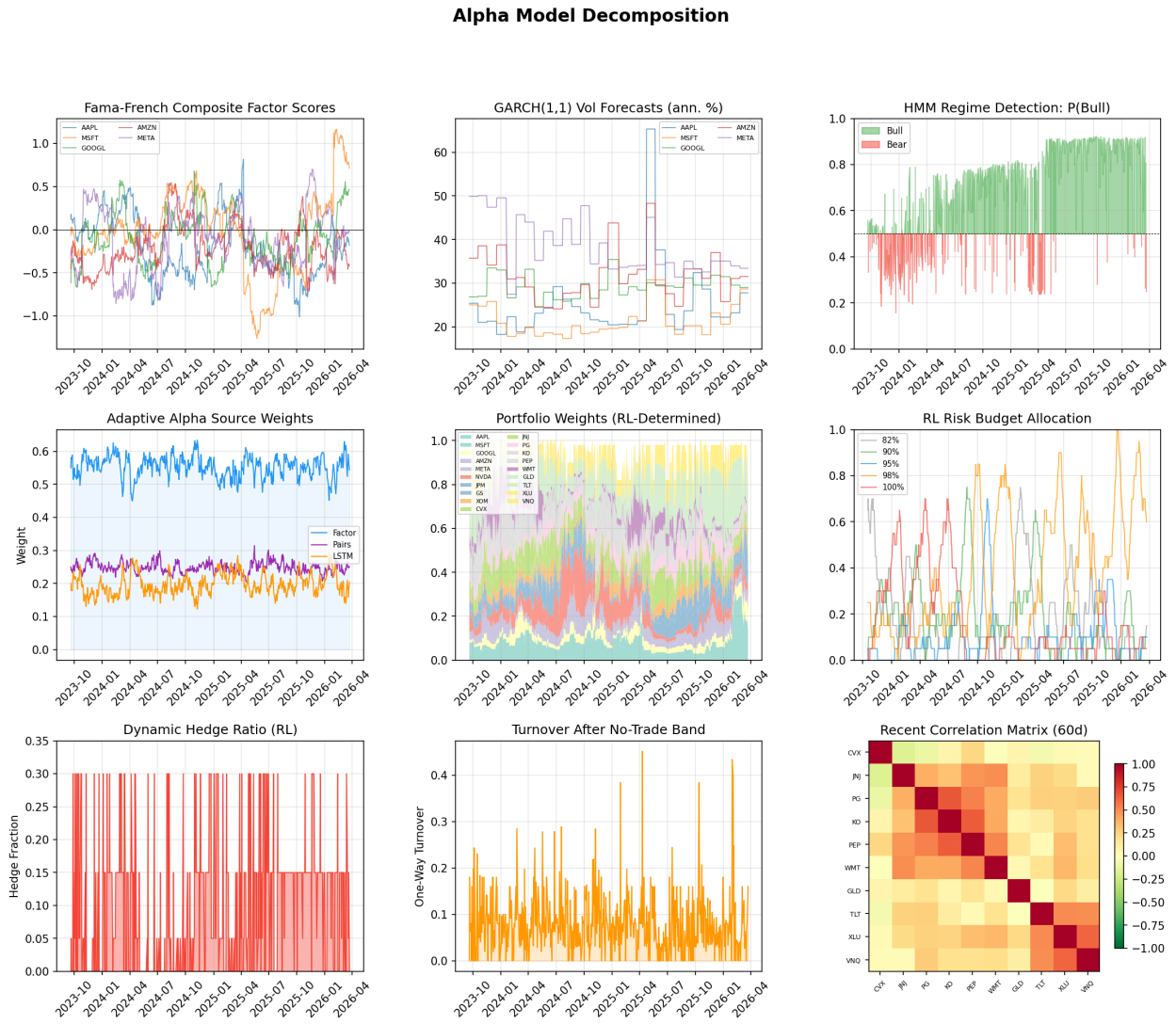


Figure 4: Alpha model diagnostics: per-sleeve signal quality and adaptive combination weights over the test window.

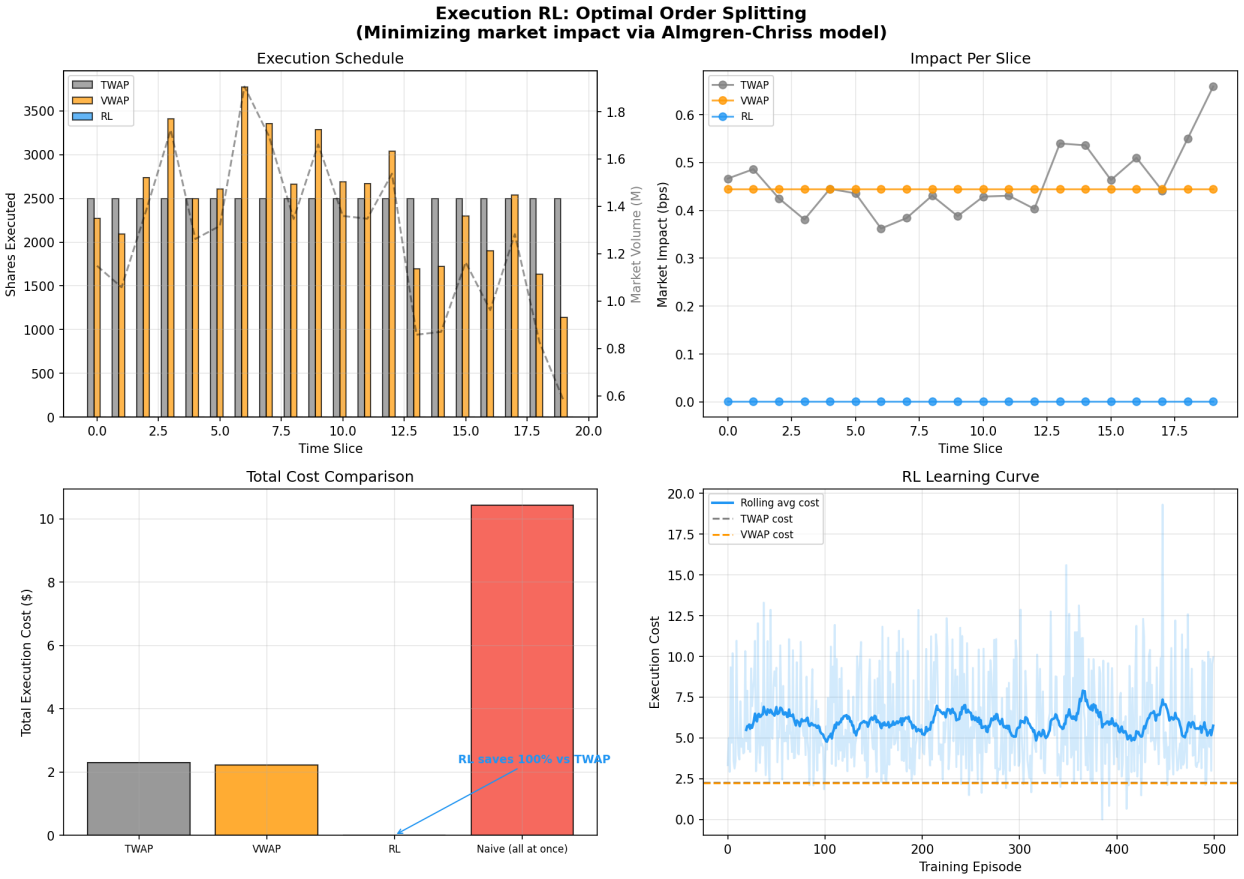


Figure 5: Execution RL layer: turnover management, no-trade band activations, and realized transaction-cost drag.

9 Discussion

The experimental results support two conclusions of particular methodological importance.

First, the data and alpha side of the pipeline is doing real work. The factor-only benchmark is strong, and equal-weight also performs well, suggesting that the broader universe and adaptive alpha combination add meaningful signal.

Second, the latest factor-anchored RL design improves the role of the control layer. Earlier versions were too defensive and suppressed upside. The current version behaves more like an institutional overlay: it keeps the strategy close to the factor book, but uses RL to modulate exposure and hedge intensity when the market state becomes less favorable.

This distinction is central. If the RL layer were treated as a second stock selector, it would risk duplicating the alpha problem with a less structured model and less data efficiency. By anchoring the policy to the factor book, the pipeline preserves interpretability: alpha determines *what* to own, while RL primarily determines *how much* to own and *how much protection* to carry.

The results also suggest a useful practical lesson. Improvements in data quality and signal specification are likely to matter more from this point onward than adding more control complexity. The current architecture is already rich enough to express nontrivial risk management behavior.

Future gains will likely come from more realistic point-in-time fundamentals, macro release lags, cross-asset regime features, and richer hedging instruments.

10 Limitations and Future Work

The implementation remains research code, and several caveats apply. Macro inputs are not yet fully vintage-aware or release-lagged, and SEC enrichment does not yet constitute a true point-in-time fundamentals database. Transaction costs remain stylized rather than venue-specific. The hedge sleeve approximates convex protection without live options data, and the evaluation still reflects a single realized market path rather than a distribution of out-of-sample windows.

Natural next steps include point-in-time fundamentals, release-lagged macro features, cross-asset regime instruments, and a more realistic hedge sleeve using options or implied-volatility information.

An additional research direction would be to evaluate robustness across rolling subperiods instead of relying on a single aggregate test window. That would make it easier to distinguish structural improvements from favorable alignment with a particular market regime.

11 Conclusion

This project demonstrates that a financially grounded hybrid architecture can be both modular and effective. Classical quantitative models remain responsible for alpha discovery, while reinforcement learning acts as a sequential risk and allocation controller. The resulting system is not simply a benchmark chaser; it is a structured research pipeline whose behavior is interpretable, testable, and extensible.

A Implementation Snapshot

A.1 Current RL Control Ladders

Portfolio action	Invested fraction	Active overlay fraction
State 0	82%	5%
State 1	90%	8%
State 2	95%	12%
State 3	98%	16%
State 4	100%	20%

Hedge action	Hedge ratio
State 0	0%
State 1	3%
State 2	8%
State 3	15%

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